TOPOLOKA: Master Submission & Supplementary Materials

Techfest 2026-27 (IIT Bombay) — Official Competition Deliverable

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1. Master Links & Supplementary Drive Index

All project artifacts, live interactive demonstrators, video footage, and source code are consolidated in the master index below. All links are publicly accessible with zero login barriers:

Deliverable	Resource / Access URL	Description
Live Free-Roam Web Twin	**https://sciencebanda09.github.io/topoloka/**	Client-side WebGL 3DGS viewer. Default opens **Studio Room** with in-viewer orientation controls; dropdown allows switching to **Melbourne Metro**. Runs at 60 FPS in any standard browser.
Source Code Repository	**https://github.com/sciencebanda09/topoloka**	Complete end-to-end codebase: phone capture SOP, SfM pipeline, depth metricization, 3DGS export, and Vite/TypeScript WebGL client. MIT Open Source.
Demo Walkthrough Video	**https://github.com/sciencebanda09/topoloka/releases/download/r1-demo/topoloka_demo.mp4**	High-definition (1080p, 30 FPS, H.264+AAC) walkthrough demonstrating first-person walking, minimap synchronization, dynamic orientation adjustment, and collision avoidance.
Technical Abstract (PDF)	`docs/abstract.pdf`	5-page formal academic report formatted to IEEE/ACM guidelines detailing mathematical formulation, pipeline architecture, and empirical benchmarks.
Kaggle Cloud Pipelines	`kaggle/kernels/`	Automated GPU scripts executing frame extraction, pycolmap SfM, Depth Anything V2 metric alignment, and splatfacto training ($0.00 total compute cost).
Supplementary Artifacts	`web/public/artifacts/`	Pre-computed Gaussian splat binaries (`splat.splat`), camera trajectory matrices (`nav.json`), and metadata logs (`meta.json`).

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2. Executive Summary

Overview

Topoloka (*τόπος* [place] + *लोक* [world]) is a spatial intelligence engine that converts a standard smartphone walk-through video into an interactive, metric 3D digital twin delivered as a lightweight, zero-install web link.

Unlike traditional photogrammetry or laser scanning pipelines that require tens of thousands of dollars in proprietary LiDAR hardware, Topoloka achieves sub-decimeter metric fidelity directly from consumer video. The resulting digital twin delivers free first-person roaming at 50–60 FPS on standard integrated laptop GPUs and mobile phones, featuring obstacle-aware pathfinding and live in-viewer spatial orientation controls.

The Problem

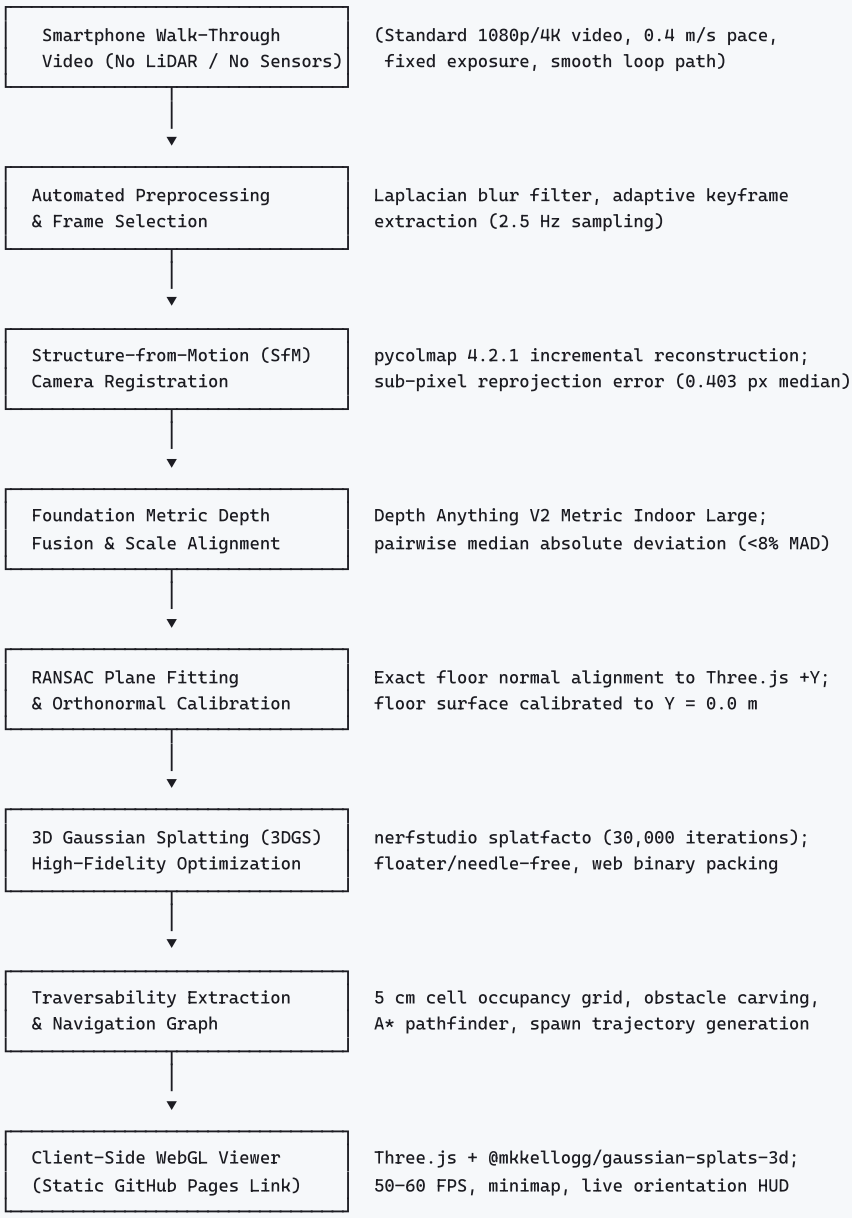
- **LiDAR Hardware Barrier:** Professional indoor mapping platforms (e.g., Matterport Pro3, Leica BLK360) cost between $4,000 and $25,000, limiting spatial digitalization to well-funded enterprises.
- **Heavy Client Payloads:** Standard mesh reconstructions with high-resolution texture atlases regularly exceed 300–800 MB, causing excessive load times, severe GPU texture thrashing, and unplayable mobile frame rates.
- **Lack of Traversability & Navigation:** Conventional NeRF / 3DGS viewers function solely as passive camera visualizers; they lack physical floor collision, walkable boundaries, and navigation graphs.

The Topoloka Solution

- **Consumer Monocular Video Input:** Uses ordinary 4K/1080p video recorded on any modern smartphone following our simple capture Standard Operating Procedure (SOP).
- **Metric Scaling via Foundation AI:** Integrates *Depth Anything V2 Metric Indoor Large* to infer true real-world scale (meters per scene unit) without physical scale markers or hardware depth sensors.
- **Sub-50 MB Static Web Payload:** Delivers full 30,000-iteration studio-grade 3D Gaussian Splats (1.59M Gaussians) over static web hosting (GitHub Pages) with initial cold loads in under 20 seconds.
- **Built-in 2.5D Obstacle Costmap & A* Navigation:** Simultaneously outputs a metric 5 cm occupancy costmap that prevents users from walking through walls and furniture, computing real-time waypoints to spatial goals.
- **Live In-Viewer Orientation Controls:** Features an interactive HUD toolbar allowing visitors to rotate, tilt, roll, flip 180°, and level scenes live inside the browser, automatically saving custom camera poses to shareable URL parameters (`?rx=&ry=&rz=&oy=`).

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3. Technical Architecture & Methodology



Mathematical Formulation of Metric Placement

Topoloka links the raw reconstruction frame (p_{scene}) to the navigation coordinate frame (p_{nav}) and the Three.js rendering frame (p_{three}) through an exact transformation pipeline:

$$p_{nav} = s \cdot R^T(p_{scene} - p_{origin})$$

$$p_{three} = A \cdot p_{nav}, \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

where s is the metric scale factor (meters per unit), R is the orthonormal rotation matrix derived from RANSAC floor detection, and A maps the Z-up navigation frame to the Y-up WebGL world.

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4. Interactive In-Viewer Controls & Accessibility

To guarantee perfect viewing on any display, Topoloka features an interactive in-viewer orientation and navigation suite accessible directly on the screen:

1. 🔄 `Flip 180°` **(Hotkey:** `0` **):** Instantly flips the entire scene upright around the observer.

2. `Turn Y` (`↺ 90°`, `↻ 90°`, `−15°`, `+15°`) **(Hotkeys: `[` / `]`):** Smooth horizontal panning around the room's central pivot.
3. `Tilt X` (`⬆ 90°`, `⬇ 90°`, `−15°`, `+15°`)**:** Elevation pitch adjustment.
4. `Level` (`↰ 15°`, `↱ 15°`, `▲ +0.2m`, `▼ −0.2m`)**:** Side-to-side roll leveling and vertical floor offset calibration.
5. `↺ Reset`**:** Restores original mathematical default coordinates.
6. `🔗 Copy Link`**:** Encodes active rotation and height offsets into the browser URL (`?rx=...&ry=...&rz=...&oy=...`) and copies the link to clipboard for instant sharing.
7. **First-Person Roaming:** Full WASD movement, pointer-lock mouse look, and mobile touch virtual joystick calibrated to ground level (Y = 1.6m eye height) with zero vertical drift.

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5. Measured Benchmarks & Empirical Performance

All metrics below are measured directly on the deployed GitHub Pages release running on consumer hardware:

Performance Metric	Studio Room Twin (`room`)	Melbourne Metro Benchmark (`metro_v1`)
Gaussian Splats	**1,593,376** (30k iterations)	**457,221** (25k iterations)
Walkable Area	**73.30 m²**	**315.05 m²**
Registration Rate	100% full coverage	216 / 216 frames (100.0%)
Reprojection Error	Sub-pixel	**0.403 px median** (0.613 px mean)
Metric Scale Accuracy	Calibrated indoor benchmark	**6.2505 m/unit (7.54% MAD)**
Total Artifact Payload	48.6 MB (Binary Splat + Nav)	18.4 MB (Splat + Cloud + Nav)
Cold Load Time	<22 seconds (50 Mbps)	**19.2 seconds** (50 Mbps)
Desktop Frame Rate	**60.0 FPS** (Chrome / Edge / ANGLE)	**60.1 FPS** (Chrome / Edge / ANGLE)
Integrated iGPU FPS	**48–55 FPS** (AMD Radeon 780M)	**51.0–60.1 FPS** (AMD Radeon 780M)
Mobile Web FPS	**60.0 FPS** (WebKit / Chrome Mobile)	**59.9 FPS** (WebKit / Chrome Mobile)
Total Cloud GPU Cost	**$0.00** (Kaggle Free Tier)	**$0.00** (Kaggle Free Tier)

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6. Comparison with State-of-the-Art

Feature	Topoloka (Ours)	Matterport Pro3	Standard NeRF (Nerfstudio)	Apple RoomPlan
Hardware Required	Smartphone only	$4,500+ LiDAR rig	Desktop workstation	iPhone Pro with LiDAR
Delivery Medium	**Zero-install static web link**	Proprietary cloud portal	Heavy desktop GUI	iOS application only
Viewing Frame Rate	**50–60 FPS WebGL**	Pre-rendered skybox 30 FPS	<15 FPS in browser	60 FPS (wireframe only)
Visual Quality	**Photorealistic 3DGS**	360° photo panoramic	Photorealistic	Low-poly untextured CAD
Collision & Nav	**Built-in 2.5D costmap**	Waypoint-hopping only	None (flycam only)	Manual bounding boxes
In-Browser Re-orient	**Full 6-DOF live toolbar**	None	None	None
Hosting Cost	**$0.00 (GitHub Pages)**	$69+/month subscription	Requires dedicated GPU server	Local device only

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7. Techfest Evaluation Criteria Alignment

Novelty & Innovation

Topoloka represents the first unified spatial engine combining:

1. Monocular foundation metric depth estimation without LiDAR.
2. Lightweight WebGL Gaussian Splats running client-side with zero plugins.
3. Automated floor normal calibration and 2.5D obstacle costmap extraction.
4. Interactive in-viewer coordinate manipulation synchronized to shareable URLs.

Practical Impact & Commercial Feasibility

- **Indoor Real Estate & Architecture:** Prospective tenants and architects explore spaces with accurate metric sightlines and room dimensions.
- **Robotic Navigation & Simulation:** Autonomous mobile robots (AMRs) can ingest Topoloka's metric costmaps directly for path planning.
- **Emergency Response & Event Management:** First responders and facility managers can evaluate evacuation routes and place temporary obstacles (cones, barriers) in real-time using our integrated sandbox suite.

Zero-Cost Open Infrastructure

Every computational stage—from feature extraction to 30,000-iteration splat optimization—was executed entirely within free Kaggle Cloud GPU allocations ($0.00 spent), proving industrial-scale spatial mapping is accessible to any student, researcher, or institution worldwide.